

LIGHT BYTES

Blue-laser optical discs are already here, and it's time for some serious excitement!

Optical media is useful in a variety of situations—for quick backups of small amounts of data, for the sheer portability they offer (it's cumbersome to have to pull your hard disk out of the cabinet when you need to send data across!), and for their role as “data-offloaders” from our growing hard disks.

QUESTIONS TO ASK

What are the different types of optical media?

Currently, there are four formats one can buy from the market, namely CD (Compact Disc), DVD (Digital Versatile Disc), Blu-ray, and HD-DVD.

CDs are available as CD-R and CD-RW, and have a capacity of 700 MB. DVDs are available as DVD-R and DVD-RW. Within DVD, there are sub-formats such as DVD+R and DVD-R; when buying, check which format your drive supports. (These days, all drives support the common formats, but if you're buying for a somewhat older burner, be sure there won't be any compatibility issues.) A single-layer DVD has a capacity of 4.7 GB and the dual-layer DVD has a capacity of 8.4 GB.

Blu-ray is a recent entrant and has a mammoth capacity of 50 GB per dual-

layer disc. For now, 25 GB discs are available, though they're expensive, and Blu-ray drives retail at more than Rs 40,000. HD-DVD is the rival format and offers similar features, but it is still to make its way to India.

Are there compatibility problems with various DVD formats?

There are certain incompatibilities with different types of DVD media. As an example, a DVD+R/RW drive can't write a DVD-R or DVD-RW disc and vice versa.

Some brands of CD and DVD media have known compatibility problems with certain brands of CD and DVD drives. A few encounter problems while reading or burning the disc. Manufacturers provide details for these on their Web sites and in the manuals. Do some quick research about whether the media will work with your drive.

What is the recommended burning speed?

Burning a disc at a lower speed is generally preferred, as it drastically reduces the chances of creating a coaster (wasted disc). Buffer underruns used to be a problem with old PCs and burning software; nowadays, PCs are fast enough, and the burning applications have become better at handling the buffer. CD writing software also suggest burning speeds depending on the type of media inserted. While burning an audio CD or MP3 CD for playing on your audio system, use a maximum burning speed of 8x for best compatibility.

How should I store the media?

Optical media have chemically treated surfaces that enable the storing of data on to them. Exposure to extreme conditions such as heat, direct sunlight, high humidity, etc. reduces their life, and cause data corruption. Optical media requires a cool, dry place for storage.

Usage Tips

- ❑ Make use of felt marker for labelling the disc. Hard-tipped pens damage the recording surface.
- ❑ Use pouches or CD covers to store the media in a cool, dry place.
- ❑ Avoid touching the surface of the disc. The more often you touch it, the sooner it will degrade!
- ❑ While cleaning a disc, use a soft linen cloth. Do not clean the media in a circular motion: clean along the radius, moving from the centre towards the edge in straight lines.

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What To Look For

- ❑ Prefer buying branded media; they might cost a little more, but data reliability is increased.
- ❑ If you burn a lot, buy a spindle. You might not think you need all those many, but you'll realise later that it was a good deal!
- ❑ Look at the burn speed mentioned; match it with what your disc burner is capable of
- ❑ Certain brands are only compatible with certain drives; buy only compatible media. This is addressed in a little more detail later.

- ❑ Make copies of critical data at regular intervals, as discs tend to degrade over a period of time and become unreliable.

- ❑ Hold the disc with one finger in the hole at the centre.

Future Trends

The launch of Blu-ray discs has jump-started a new revolution on the optical storage media storage front. 25 GB or 50 GB on a single disc! Who would need so much space? Well, that was the same question everyone was asking when the CD-ROM came out! With HD (High Definition) content making huge inroads, the need is justified.

Availability in India of Blu-ray media is limited, and recorders are very expensive, but it's only a matter of time before we use Blu-ray and/or HD-DVD as regularly as we do CDs and DVDs today. ■